RoT: Enhancing Table Reasoning with Iterative Row-Wise Traversals

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Abstract

The table reasoning task, crucial for efficient data acquisition, aims to answer questions based on the given table. Recently, reasoning large language models (RLLMs) with Long Chain-of-Thought (Long CoT) significantly enhance reasoning capabilities, leading to brilliant performance on table reasoning. However, Long CoT suffers from high cost for training and exhibits low reliability due to table content hallucinations. Therefore, we propose Row-of-Thought (RoT), which performs iteratively row-wise table traversal, allowing for reasoning extension and reflection-based refinement at each traversal. Scaling reasoning length by row-wise traversal and leveraging reflection capabilities of LLMs, RoT is training-free. The sequential traversal encourages greater attention to the table, thus reducing hallucinations. Experiments show that RoT, using non-reasoning models, outperforms RLLMs by an average of 4.3%, and achieves state-of-the-art results on WikiTableQuestions and TableBench with comparable models, proving its effectiveness. Also, RoT outperforms Long CoT with fewer reasoning tokens, indicating higher efficiency.

1 Introduction

Table reasoning is an important task where the input consists of a question and the table, and the output is the answer based on the table (Jin et al., 2022; Zhang et al., 2025d). Tables typically comprise multiple rows, with each row containing several information-dense cells (Ruan et al., 2024). Automated table reasoning attracts considerable research interest due to its potential to extract valuable information from tables, thus accelerating data acquisition (Badaro et al., 2023; Lu et al., 2025).

Recent advancements in reasoning large language models (RLLMs) have significantly enhanced reasoning capabilities utilizing Long Chain-of-Thought (Long CoT), including table reasoning capabilities (Li et al., 2025b; Qian et al., 2025). This improvement stems from Long CoT, which sequentially scales the length of CoT, engages in self-reflection, and explores diverse reasoning paths, in contrast to the shallow and direct reasoning of Short CoT (Chen et al., 2025; Yeo et al., 2025). However, Long CoT exhibits two limitations in table reasoning, as illustrated in Figure 1 (a): (*i*) **High Cost**: Achieving Long CoT capabilities for improved table reasoning capabilities necessitates high-quality data, leading to substantial training expenses (Qian et al., 2025; Jiang et al., 2024a). (*ii*) **Low Reliability**: As the output reasoning chains lengthen, models are prone to losing relevant tabular information from the input, resulting in hallucinations of the tabular content (Zhang et al., 2023; Liu et al., 2025a,b; Kumar et al., 2025).

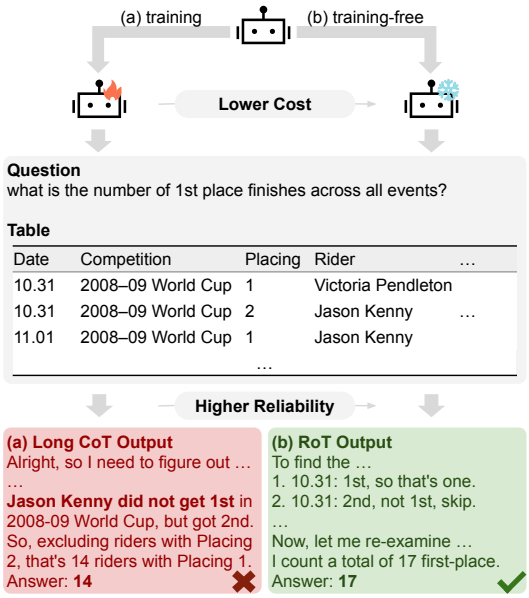


Figure 1: Compared with (a) Long CoT, (b) RoT necessitates no training, exhibits lower costs, and enhances reliability by mitigating hallucination via sequentially row-wise table traversal.

Therefore, we propose **R**ow-**o**f-**T**hought (RoT),

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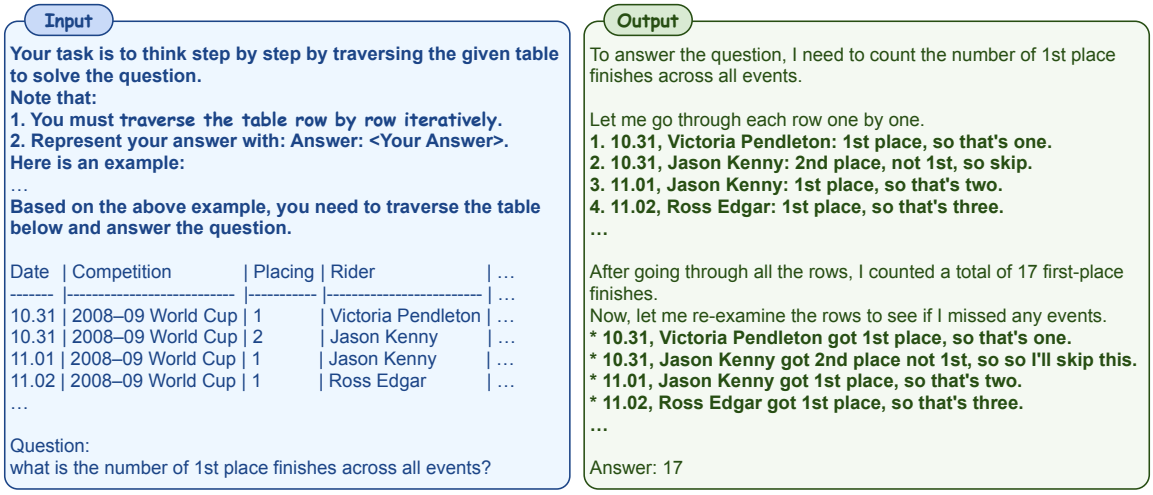


Figure 2: The overview of RoT with the input and output of the example. The instruction is highlighted with **blue** and the iterative row-wise table traversal process is highlighted with **green**.

a novel method that enhances table reasoning by guiding the model to perform iteratively row-wise traversal reasoning, as illustrated in Figure 1 (b). Row-wise traversal refers to the reasoning process where it considers information from a single row at each step to update intermediate results. In the iterative process, after each traversal, the model can either extend its reasoning or reflect on prior steps and initiate a new traversal accordingly. RoT alleviates two limitations of Long CoT: (*i*) Low Cost: Since RoT sequentially scales the reasoning length by row-wise traversals and the self-reflection capabilities are equipped in LLMs (Gu et al., 2025; AI et al., 2025), RoT is training-free and can be implemented with non-reasoning large language models (non-RLLMs) through prompting. (*ii*) High Reliability: By prompting the sequential traversal of all rows, RoT directs greater attention to tabular information thoroughly, thereby mitigating hallucination (Shi et al., 2024a; Chuang et al., 2024).

To demonstrate the effectiveness of RoT, we conduct experiments on WikiTableQuestions (Pasupat and Liang, 2015), HiTab (Cheng et al., 2022), and TableBench (Wu et al., 2024). Compared to Long CoT on RLLMs, RoT achieves an average improvement of 4.3% with non-RLLMs without training, validating its effectiveness. Furthermore, RoT can also enhance the performance of RLLMs with an average improvement of 2.4%, mitigating their table content hallucination. Additionally, RoT achieves state-of-the-art (SOTA) results on WikiTableQuestions and TableBench with comparable models, and yields competitive results on HiTab. Analysis experiments reveal that **RoT with non-RLLMs outperforms Long CoT with fewer reasoning tokens, showing higher efficiency**.

Our contributions are as follows:

1. We propose RoT, which achieves lower cost without training and higher reliability compared to Long CoT.
2. RoT on non-RLLMs outperforms Long CoT on RLLMs by an average of 4.3% and achieves SOTA results among comparable models on WikiTableQuestions and TableBench, proving its effectiveness.
3. RoT with non-RLLMs outperforms Long CoT using fewer reasoning tokens, highlighting its higher efficiency.

2 RoT

To mitigate the limitations of High Cost and Low Reliability in Long CoT, we propose RoT. As illustrated in Figure 2, RoT enhances table reasoning capabilities by iterative row-wise traversals. The complete prompts are available in Appendix A.1.

2.1 Overview

Given an instruction I, a question Q, a table U composed of M rows and N columns, and in-context demonstrations D, the model outputs a step-by-step reasoning process that iteratively traverses the table in the sequential row order until the final answer is derived. Formally, $R; A = \mathcal{F}(I, Q, U, D)$, where $\mathcal{F}$ is the LLM, and $R; A$ denotes the concatenation of the reasoning process R and the answer A. We represent the table in Markdown format, following previous works (Wang et al., 2024; Zhang et al., 2024b; Yu et al., 2025). We now introduce the two

key factors in the reasoning process R in RoT.

2.2 Traversal

We first detail the traversal reasoning adopting the row as the traversal unit in RoT. Specifically, the model assesses the relevance of information within the current row and infers intermediate results according to the question and prior inference. Formally, $R_i; A_i = R_{i,1}; A_{i,1}, R_{i,2}; A_{i,2}, ..., R_{i,M}; A_{i,M}$. R_i represents the reasoning process of the i-th traversal, and A_i is the result obtained in the i-th traversal. $R_{i,j}$ denotes the reasoning over the j-th row of the table during the i-th traversal, and $A_{i,j}$ is the corresponding intermediate result. RoT leverages the inherent structural features of tables by decomposing the problem-solving into fine-grained, step-by-step reasoning, with each step corresponding to a row. By accumulating the intermediate results $A_{i,j}$ from each row, we obtain the result A_i after one traversal. The row-wise traversal not only brings the reasoning length scaling but also mitigates hallucination by forcing the model to attend to the entire table content. We also discuss comparisons with adopting other traversal units and RoT in §3.4.5.

2.3 Iteration

The iteration process allows the model to continue reasoning after a traversal, which is necessary for multi-hop questions that cannot be answered in a single traversal. Also, the model can choose to reflect on the previous reasoning after a traversal and subsequently revisit the table based on the reflection until the final answer is obtained. Formally, the iterative reasoning process can be represented as $R; A = R_1; A_1, R_2; A_2, ..., R_T; A_T$, where T is the total number of traversals. Rather than predefining T in the prompt, the model dynamically decides to terminate inference when the final answer has been obtained. We provide a detailed analysis of the iterative table traversals in §3.4.2. We also provide case study for iterative traversals in Appendix C.2.

3 Experiments

3.1 Experimental Setup

Dataset RoT is evaluated on three widely used table reasoning datasets: WikiTableQuestions (Pasupat and Liang, 2015), HiTab (Cheng et al., 2022), and TableBench (Wu et al., 2024), following previous works (Jiang et al., 2024b; Cao, 2025; Li et al., 2025a). WikiTableQuestions is a mainstream table-based question answering dataset. HiTab focuses on hierarchical tables, challenging models to comprehend complex structural relationships. TableBench presents a challenging benchmark covering diverse question types and topics.

Models (*i*) For **non-RLLMs**, we utilize Llama3.1-8B-Instruct (Llama3.1-8B), Llama3.3-70B-Instruct (Llama3.3-70B) (Dubey et al., 2024), Qwen2.5-7B-Instruct (Qwen2.5-7B), and Qwen2.5-32B-Instruct (Qwen2.5-32B) (Yang et al., 2024a). (*ii*) For **RLLMs**, we employ the corresponding-sized DeepSeek-R1-Distill-Llama-8B (R1-Llama-8B), DeepSeek-R1-Distill-Llama-70B (R1-Llama-70B), DeepSeek-R1-Distill-Qwen-7B (R1-Qwen-7B), and DeepSeek-R1-Distill-Qwen-32B (R1-Qwen-32B) (Guo et al., 2025). We exclude Qwen2.5-Math-7B, which is the base model of R1-Qwen-7B, due to its primary focus on solving mathematical tasks, resulting in suboptimal performance on the table reasoning task (Yang et al., 2024b).

Metric For WikiTableQuestions and HiTab, we adopt accuracy as the evaluation metric, following prior works (Pasupat and Liang, 2015; Cheng et al., 2022). Accuracy measures the ability of models to generate answers that exactly match the gold answers. For TableBench, we use Rouge-L (Lin, 2004), consistent with the previous research (Wu et al., 2024). Rouge-L evaluates the quality of generated answers based on the longest common subsequence, considering both precision and recall.

Baselines RoT employs the one-shot and zero-shot prompts to enable **non-RLLMs** and **RLLMs** to perform iterative row-wise traversals, respectively (prompts in Appendix A.1). We do not use demonstrations for RLLMs due to the performance degradation using the few-shot prompt observed in Appendix B.1 (Guo et al., 2025; Zheng et al., 2025). We compare RoT with the following methods:

- Short CoT: We prompt **non-RLLMs** to engage in step-by-step reasoning with the one-shot prompt, which uses the same demonstration as RoT.
- Long CoT: We utilize the zero-shot prompt for **RLLMs**.
- Previous table reasoning works: We compare RoT with existing table reasoning methods with comparable models.

Model	Method	WikiTQ	HiTab	TableBench
Llama3.1-8B (Dubey et al., 2024)	Short CoT	57.9	46.5	31.5
	RoT	63.6 (+2.7)	**56.6** (+10.1)	**35.7** (+4.2)
R1-Llama-8B (Guo et al., 2025)	Long CoT	62.7	49.7	34.9
	RoT	**63.7** (+1.0)	50.9 (+1.2)	35.4 (+0.5)
Llama3.3-70B (Dubey et al., 2024)	Short CoT	72.7	66.9	38.2
	RoT	**78.7** (+6.0)	**72.4** (+5.5)	**44.8** (+6.6)
R1-Llama-70B (Guo et al., 2025)	Long CoT	76.2	67.4	40.4
	RoT	78.3 (+2.1)	68.6 (+1.2)	42.8 (+2.4)
Qwen2.5-7B (Yang et al., 2024a)	Short CoT	52.2	54.7	30.9
	RoT	**61.7** (+9.5)	**58.9** (+4.2)	34.9 (+4.0)
R1-Qwen-7B (Guo et al., 2025)	Long CoT	53.3	50.2	34.2
	RoT	57.1 (+3.8)	51.2 (+1.0)	**35.6** (+1.4)
Qwen2.5-32B (Yang et al., 2024a)	Short CoT	69.2	70.3	35.9
	RoT	75.6 (+6.4)	**76.6** (+6.3)	40.4 (+4.5)
R1-Qwen-32B (Guo et al., 2025)	Long CoT	69.6	70.8	38.0
	RoT	**76.9** (+7.3)	73.5 (+2.7)	**42.0** (+4.0)

Table 1: Performance comparison between RoT and baselines, where WikiTQ and HiTab use accuracy as the evaluation metric and TableBench uses Rouge-L. WikiTQ refers to WikiTableQuestions. For each dataset, the highest performing result among models of the same scale is **bolded**. Performance gain compared to baselines is highlighted with (green).

Dataset	Previous SOTA	RoT
WikiTQ	78.0 (Cao, 2025)	**78.7**
HiTab	**79.1** (Jiang et al., 2024b)	76.7
TableBench	43.9 (Wu et al., 2024)	**44.8**

Table 2: Performance comparison between RoT and SOTA methods with similar scale models.

3.2 Main Results

Table 1 presents a comparison between RoT and baselines using different models across datasets. RoT, using non-RLLMs consistently and significantly outperforms Long CoT with RLLMs, achieving an average improvement of 4.3%, demonstrating its effectiveness. Furthermore, RoT yields an average increase of 2.4% in the performance of RLLMs, indicating its effectiveness in mitigating the limitations of Long CoT. We also observe that:

RoT outperforms baselines consistently. RoT surpasses Long CoT primarily because it enforces the row-wise traversals, alleviating hallucinations in Long CoT (Zhang et al., 2023; Shi et al., 2024a; Liu et al., 2025b). Compared to Short CoT, RoT achieves superior performance through fine-grained, row-wise reasoning, thereby reducing the complexity of individual reasoning steps and minimizing the risk of overlooking relevant details (Snell et al., 2024; Wang et al., 2024).

We also compare RoT with SOTA methods on three datasets, as shown in Table 2. Due to space constraints, detailed comparisons with prior works are provided in Appendix B.2. RoT gets SOTA results on WikiTQ and TableBench and is comparable with the SOTA method on HiTab, highlighting its effectiveness. The comparable performance on HiTab can be attributed to the fact that RoT does not incorporate specific enhancements for hierarchical tables, unlike previous methods (Zhao et al., 2023; Jiang et al., 2024b; Li et al., 2025a).

RoT improves performance across varying models. RoT significantly enhances the table reasoning capabilities of various non-RLLMs and RLLMs without training. RoT with RLLMs does not outperform RoT with non-RLLMs consistently because, while we mitigate hallucination in Long CoT, they exhibit problems such as overthinking, which are less pronounced in non-RLLMs (Yin et al., 2025; Zeng et al., 2025). Additionally, R1-Qwen-7B does not outperform Qwen2.5-7B on HiTab, as its base model, Qwen2.5-Math-7B, is optimized for mathematical reasoning, unlike the general base models of others (Yang et al., 2024b).

3.3 Ablation Experiments

To demonstrate the effectiveness of RoT, we conduct ablation experiments on three datasets, as shown in Table 3. The prompts used in the ablation experiments are provided in Appendix A.2.

Scale	Model	Method	WikiTQ	HiTab	TableBench
8B	Llama3.1	RoT	**63.6**	**56.6**	**35.7**
		w/o Iteration	60.7	55.5	32.7
		w/o Traversal	55.2	42.2	31.2
	R1-Llama	RoT	**63.7**	**50.9**	**35.4**
		w/o Iteration	56.6	48.9	31.5
		w/o Traversal	46.8	36.8	25.7

Table 3: The ablation results of RoT compared with reasoning with one single table traversal (denoted as *w/o Iteration*) and reasoning without table traversal (denoted as *w/o Traversal*). For each dataset, the highest performing result with the same model is **bolded**.

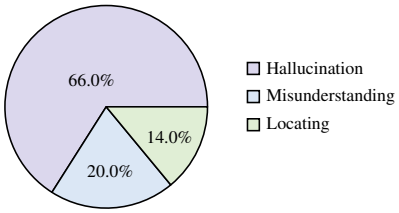


Figure 3: Long CoT underperforms RoT due to the error types, with their distribution.

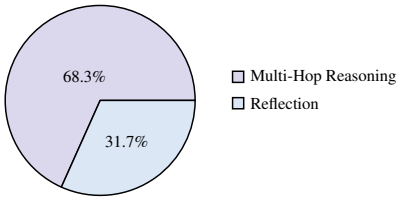


Figure 4: The distribution of reasons for iterative traversals in RoT on sampled 60 instances from three datasets.

Effectiveness of Iteration To validate the effectiveness of iterative reasoning in RoT, we prompt the model to perform only a single table traversal. The results indicate a consistent performance decrease compared to RoT when iteration is removed, demonstrating that iterative traversal effectively aids the model in exploration and reflection. Also, a single traversal is insufficient to adequately address all table reasoning questions.

Effectiveness of Traversal To demonstrate the importance of traversal in RoT, we prompt LLMs to iteratively reflect instead of iteratively traversing the table. The significant performance decline observed underscores that traversing the table, through scaling reasoning length and mitigating hallucinations of tabular content, effectively enhances table reasoning.

3.4 Analysis Experiments

We primarily select Llama3.1-8B and R1-Llama-8B for subsequent analysis experiments due to their high reasoning efficiency and space limitations.

3.4.1 Why RoT Outperforms Long CoT?

To explore the superior performance of RoT over Long CoT, we conduct an error analysis on WikiTQ instances where RoT with Llama3.1-8B succeeds while Long CoT with R1-Llama-8B fails. We also explore why RoT with RLLMs outperforms Long CoT in Appendix B.6. Figure 3 illustrates the identified error categories on sampled 50 instances, which are detailed below. We provide the cases of each error category in Appendix C.1.

(*i*) **Hallucination** refers to the model incorrectly recalling tabular information, leading to inconsistencies between the table input and the generated reasoning, such as cross-row confusion and relevant information omission. Long CoT suffers from severe hallucinations, primarily due to the increasing loss of tabular content as the reasoning chain lengthens (Liu et al., 2025b). Conversely, RoT performs row-wise traversals sequentially, guides greater attention to the table content, which mitigates this issue (Yin et al., 2020; Badaro et al., 2023). (*ii*) **Misunderstanding** denotes the misinterpretation of the question, which is a common challenge for distilled models (Banerjee et al., 2024; Yin et al., 2025). (*iii*) **Locating** refers to incorrectly identifying the relevant table location for the given question. Therefore, RoT demonstrates a higher reliability compared to Long CoT.

3.4.2 How does the number of traversals affect RoT?

To examine when RoT requires iterative traversals, we randomly select 20 instances from each dataset on Llama3.1-8B where RoT traverses the table more than once and investigate the reasons, as shown in Figure 4. We provide a detailed explanation of the reasons below, with examples provided in Appendix C.2. (*i*) **Multi-Hop Reasoning**: The

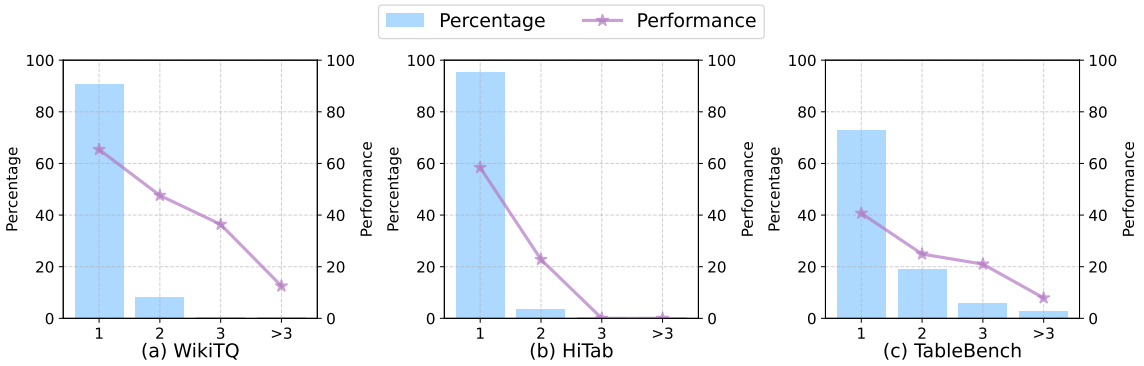


Figure 5: The distribution of table traversal counts and the corresponding performance of RoT on three datasets with Llama3.1-8B.

inherent complexity of certain questions demands iterative table traversals to derive the solution, particularly when addressing *cross-row dependencies*. (*ii*) **Reflection**: The model reflects on its prior reasoning upon completing a traversal and initiates new reasoning passes accordingly. This demonstrates that RoT with non-RLLMs equips the capacities of extending reasoning and self-reflection on table reasoning.

Additionally, to assess the impact of traversal count on performance, we report the distribution of table traversal counts and the corresponding performance when using Llama3.1-8B, as depicted in Figure 5. We observe that: (*i*) On the more challenging TableBench dataset, RoT tends to perform more traversals as required. (*ii*) Increasing traversal counts correlate with a decrease in the performance of RoT, due to the inherent difficulty of questions necessitating iterative traversals and the potential for exceeding token limits during such processes.

3.4.3 How does reasoning length affect table reasoning capabilities?

To investigate the impact of reasoning length on table reasoning performance, we calculate the average number of tokens used in correct and incorrect reasoning on WikiTQ, as shown in Figure 6. The results reveal that:

(*i*) RoT with non-RLLM achieves improved table reasoning with fewer tokens compared to Long CoT, demonstrating its efficiency. RoT allows the model to dynamically determine the number of iterations and non-RLLMs are not specifically trained on Long CoT data, therefore, RoT mitigates overthinking prevalent in Long CoT (Yin et al., 2025). Additionally, when using the same RLLM, RoT exhibits shorter incorrect reasoning compared to Long CoT, since RoT, by focusing more intently on the table, reduces model hallucinations regarding table content, thereby decreasing the frequency of ineffective reflections, as discussed in Appendix B.6 (Shi et al., 2024a; Qin et al., 2025). (*ii*) Using the same model, RoT produces longer correct reasoning compared to its corresponding CoT baseline. This is because the row-wise table traversal enables more fine-grained reasoning, leading to increased reasoning length and improved performance (Qian et al., 2025).

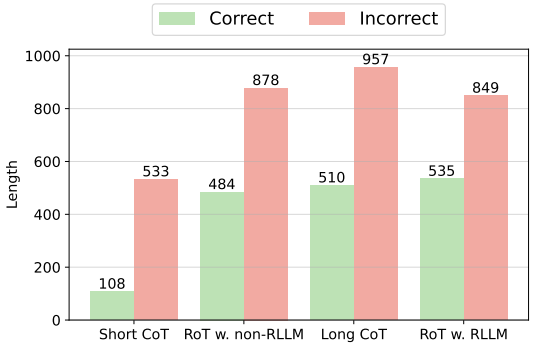


Figure 6: Comparison of average reasoning lengths for correct and incorrect inferences across three datasets on WikiTQ with Llama3.1-8B (denoted as w. non-RLLM) and R1-Llama-8B (denoted as w. RLLM).

3.4.4 How does RoT change with table size?

To evaluate the performance of RoT relative to baselines across varying table sizes, we analyze the performance of Llama3.1-8B and R1-Llama-8B on tables of different sizes in WikiTQ, defined as the product of the number of rows and columns (Figure 7). The key observations are as follows: (*i*) Overall, RoT outperforms the baselines across table sizes. (*ii*) While exhibiting a general downward trend, the performance of RoT demonstrates relative stability with increasing table size. The row-wise traversals could lead to exceeding the to-

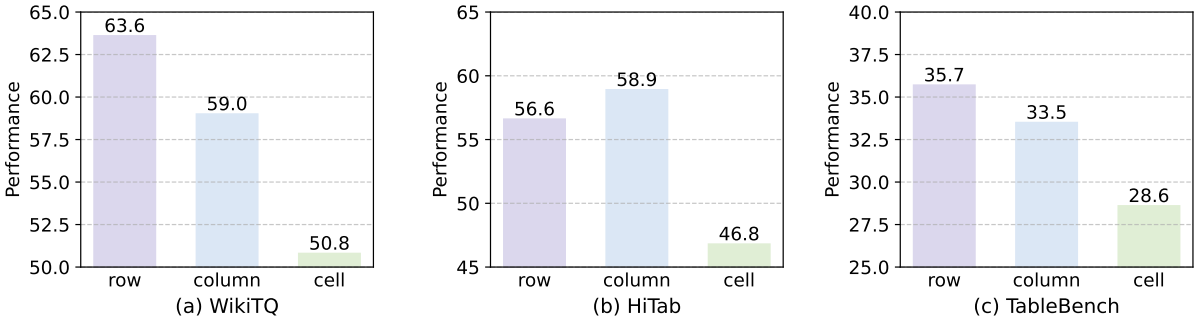


Figure 8: Comparison of RoT traversing the table with different units across three datasets.

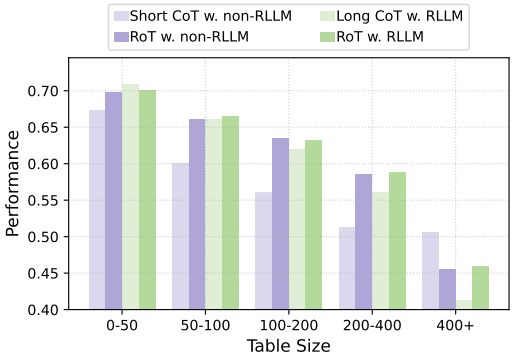


Figure 7: The comparison of the average performance of RoT and baselines on different table sizes in WikiTQ with Llama3.1-8B and R1-Llama-8B.

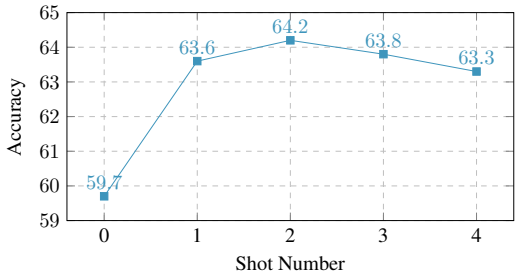


Figure 9: Performance of RoT on WikiTQ with varying numbers of demonstrations.

ken limit when the number of rows becomes excessively large before a response is generated. Long CoT suffers from an increased number of reasoning steps with larger tables, elevating the risk of hallucinating relevant information and surpassing token limits more significantly (Zeng et al., 2025; Sui et al., 2025). Short CoT, while less susceptible to token limit issues, could overlook relevant table information due to its coarser reasoning granularity and miss self-reflection reasoning (Snell et al., 2024; Zhang et al., 2025b).

3.4.5 How does the traversal unit affect RoT?

To investigate the effect of traversal units on RoT, we conduct experiments using rows, columns, and individual cells as traversal units across three datasets with Llama3.1-8B. Row-wise traversal is adopted as the default setting in the main experiments. The results indicate the following:

(*i*) On WikiTQ and TableBench, row-wise traversal achieves the best performance. Compared to column-wise traversal, row-wise traversal better aligns with the attention mechanism, enabling more effective focus on all cells within the same row (Yin et al., 2020; Liu et al., 2024a). Cell-wise traversal resulted in a significant performance decrease, due to its overly fine-grained reasoning granularity and the presence of numerous irrelevant cells, which introduce redundant reasoning steps and increase the risk of error accumulation (Jin et al., 2024; Chen et al., 2024; Patnaik et al., 2024). (*ii*) Column-wise traversal yields superior performance on HiTab. In HiTab, all tables include hierarchical row headers, while hierarchical column headers are present in 93.1% of the tables, a relatively less frequent occurrence (Cheng et al., 2022). Consequently, each cell in a row corresponds to hierarchical row headers. During row-wise traversals, each cell should be mapped to multiple row headers, whereas column-wise traversals inherently incorporate header information into each column, facilitating more effective reasoning (Zhao et al., 2023).

3.4.6 How does the number of demonstrations affect RoT?

To investigate the effect of the number of demonstrations on RoT, we conduct experiments on WikiTQ using Llama3.1-8B, as illustrated in Figure 9. All demonstrations were sampled from the WikiTQ training set. We observe that: (*i*) A substantial performance gain is observed when transitioning from zero-shot to one-shot prompting. This suggests that

a single demonstration significantly aids the model in comprehending the instruction and replicating the reasoning process for iterative row-wise table traversals, thus improving table reasoning capabilities. (*ii*) With a further increase in the number of demonstrations, performance initially improves but subsequently declines. A limited number of demonstrations is sufficient for the model to understand the instructions and learn the reasoning patterns. Additional demonstrations contribute little new information and may constrain the reasoning paths (Lin et al., 2024; Wan et al., 2025; Zheng et al., 2025). The one-shot prompt is chosen for our main experiments, balancing competitive performance with excellent inference efficiency.

4 Related Works

4.1 Table Reasoning

The table reasoning task, which aims to answer user queries through inference over tabular data, is extensively applied in data-intensive domains such as finance and research (Jin et al., 2022; Zhang et al., 2025d). Leveraging large language models (LLMs) has emerged as a prevalent method for table reasoning (Chen, 2023; Lu et al., 2025). To enhance the table reasoning capability, researchers propose to collect or augment tabular data for fine-tuning (Zhang et al., 2024a, 2025c; Su et al., 2024). However, the resource demands and potential reduction in generalization (Deng and Mihalcea, 2025) motivate training-free methods. Some methods focus on question decomposition to mitigate reasoning complexity (Ye et al., 2023; Wu and Feng, 2024; Jiang et al., 2024c). For instance, TID (Yang et al., 2025) extracts triples from the question and transforms them into sub-questions for comprehensive decomposition. Another direction involves the integration of programs or tools to facilitate reasoning (Jiang et al., 2023; Shi et al., 2024b; Zhang et al., 2024c), exemplified by MACT (Zhou et al., 2025), which employs a planning agent and a coding agent to select appropriate actions and tools for reasoning.

Recent advancements in RLLMs demonstrate that the integration of Long CoT significantly improves their reasoning abilities, including table reasoning (Chen et al., 2025; Qian et al., 2025). However, Long Long CoT suffers from significant tabular content hallucination (Zeng et al., 2025). To address this, we propose an iteratively row-wise traversal method, which mitigates hallucination by forcing the model to focus on tabular content.

4.2 Long CoT

RLLMs, such as OpenAI O1 (OpenAI et al., 2024a) and DeepSeek R1 (Guo et al., 2025), significantly improve reasoning capabilities by incorporating Long CoT with scaling reasoning length and iterative exploration and reflection, leading to consistent performance gains across diverse tasks (Snell et al., 2024; Aggarwal and Welleck, 2025). RLLMs are typically derived from base LLMs through supervised fine-tuning (SFT) or reinforcement learning (RL) (Chen et al., 2025; Chu et al., 2025). SFT aims to replicate sophisticated reasoning patterns from human-annotated or distilled data (Trung et al., 2024; Wen et al., 2025). For instance, s1 (Muennighoff et al., 2025) and LIMO (Ye et al., 2025) enhance their reasoning abilities through SFT by collecting $1,000$ and 817 high-quality training instances with meticulously labeled rationales, respectively. RL further refines reasoning abilities through self-learning and preference optimization (Liu et al., 2024b; Xu et al., 2025). For example, Zhang et al. (2025a) proposes a Process-based Self-Rewarding paradigm, which fine-tunes models using synthesized step-wise preference data.

However, previous works require high-quality data and exhibit significantly high cost (Jiang et al., 2024a; Qin et al., 2024). Given that table reasoning tasks involve structured evidence, we propose RoT that enhances the table reasoning capabilities of non-reasoning LLMs without training.

5 Conclusion

Considering the limitations of Long CoT on the table reasoning task, we focus on enhancing table reasoning capabilities with low cost and high reliability. Specifically, we propose a training-free method, RoT, which prompts the model to perform iterative row-wise traversal reasoning until the final answer is obtained. Experiments show that RoT, using non-RLLMs, outperforms Long CoT with RLLMs, achieving an average improvement of 4.3%, demonstrating the effectiveness of RoT. Additionally, RoT with RLLMs brings an average improvement of 2.4% compared with Long CoT, leading to higher reliability. Furthermore, RoT attains SOTA performance on WikiTableQuestions and TableBench among comparable models, validating its effectiveness. Analysis experiments indicate that RoT with non-RLLMs achieves better performance than Long CoT with fewer reasoning tokens, showing its higher efficiency.

Limitations

(*i*) We do not conduct experiments on multi-turn table question answering datasets. We will explore the effectiveness of RoT on such datasets in future work. (*ii*) Our experiments are exclusively performed on English datasets. We leave experimentation with RoT on different languages for future research.

Ethics Statement

All models used in this paper are publicly available, and our utilization of them strictly complies with their respective licenses and terms of use.

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A Prompts

A.1 Demonstrations of RoT

The instructions for RoT are shown in Figure 2, so in this section, we present demonstrations used across three datasets in Table 4. We select the same demonstration from the WikiTQ training set for both WikiTQ and TableBench, as the tables in these two datasets are flat. Our primary goal is to help the model understand the process of row-wise table traversals through the demonstration. In contrast, the tables in HiTab are hierarchical. Due to this distinct structure, we select the demonstration from the HiTab training set to better facilitate the understanding of the table structure.

A.2 Prompts for ablation experiments

We show the prompts used in ablation experiments in Table 5. In the ablation study, the demonstrations used are consistent with those in the main experiments, with the corresponding iterative and traversal processes removed from the reasoning process.

B Additional Experiments

B.1 Long CoT with few-shot prompt

In this subsection, we present the performance of Long CoT using few-shot prompts with R1-Llama-8B, as shown in Table 6. It can be observed that, across three datasets, the performance of Long CoT significantly declines compared to the zero-shot setting. Therefore, in the main experiments, we employ zero-shot prompts.

B.2 Comparison with previous methods

In this subsection, we present a comparison of RoT with previous works, as shown in Table 7, Table 8, and Table 9. RoT achieves state-of-the-art performance on WikiTQ and TableBench, and performs comparably to prior methods on HiTab, demonstrating its effectiveness. RoT surpasses prior methods by optimizing the table reasoning process through detailed, iterative exploration and reflection.

Notably, Table-Critic (Yu et al., 2025) introduces a multi-agent system for table reasoning, comprising a Judge to identify errors, a Critic to analyze these identified errors, a Refiner to rectify them, and a Curator to aggregate critic knowledge for enhanced critique quality. RoT surpasses Table-Critic using the same LLM, demonstrating not only effective reflection on previous reasoning but also sequential scaling through row-wise traversal, leading to improved table reasoning capabilities.

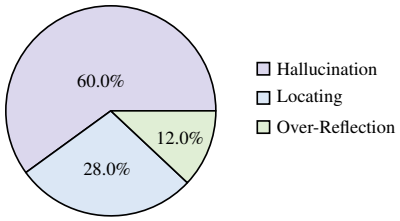


Figure 10: Long CoT underperforms RoT with RLLMs due to the error types, with their distribution.

B.3 Comparison with table-specific LLMs

We compare the performance of RoT, using Llama3.1-8B, against the table-specific LLMs. The results are shown in Table 10. As we can see, RoT demonstrates superior or comparable performance compared with LLMs finetuned on tabular data.

B.4 Robustness of RoT

To assess the robustness of RoT, including the sensibility of our termination strategy and the prompt design, we employ GPT-4o (OpenAI et al., 2024b) to paraphrase the original prompt in three different ways, while preserving the core termination logic. We then examine whether the traversal count and the performance remain consistent across these rephrased prompts. The results in Table 11 indicate that the average number of traversals across the four variants remains consistently around 1.1. This suggests that the termination behavior under RoT is robust. Also, the original prompt consistently yields the best performance. Moreover, the performance variations across different rephrasings are within 1 percentage point and remain significantly higher than the Short CoT baseline (57.9). This demonstrates the overall stability and effectiveness of our prompt.

B.5 Efficiency of RoT

Table 12 compares the number of output tokens of RoT with Long CoT across various datasets. As shown, RoT consistently outperforms Long CoT while reducing the number of reasoning tokens by 7% to 50%, demonstrating its high efficiency.

B.6 Why RoT with RLLMs outperforms Long CoT?

To analyze specifically why RoT with RLLMs outperforms Long CoT, we randomly select instances

from WikiTQ where RoT using R1-Llama-8B provided the correct answer, but Long CoT using R1-Llama-8B failed. We manually analyze the reasons for these discrepancies, with the distribution shown in Figure 10. Among them, Hallucination and Locating are as described in §3.4.1. Over-Reflection refers to cases where the reflection process led to an originally correct answer being changed to incorrect, or where excessive reflections exceeding the token limits prevented a final answer from being generated. The results indicate that: (*i*) RoT significantly mitigates the hallucination issue prevalent in Long CoT. (*ii*) The sequential row-wise traversal enhances the ability to locate all relevant information. (*iii*) RoT can alleviate Over-Reflection to some extent by guiding the reflection process through structured table traversal, thus reducing ineffective or erroneous reflections.

B.7 Why does RoT not fix the number of iterations?

Method	Performance
RoT	**63.6**
Traversal count = 1	60.7
Traversal count = 2	61.9
Traversal count = 3	59.7

Table 13: Comparison of RoT with fixing the traversal count on WikiTQ using LLaMA3.1-8B.

We conduct experiments on WikiTQ using LLaMA3.1-8B with a fixed number of traversal count. The results are summarized in Table 13. Overall, RoT achieves the best performance. In contrast, fixing the number of reasoning steps leads to a noticeable performance drop. Specifically, limiting the model to a single traversal results in a substantial degradation in accuracy, which highlights the importance of iterative reasoning in RoT. Moreover, performance with three traversals is lower than that with two, suggesting that excessive iterations may introduce unproductive reflections and consequently impair model performance.

C Case Study

C.1 Case study of RoT compared with Long CoT

We present examples where RoT outperforms Long CoT for distinct reasons, as illustrated in Figure 11, Figure 12, and Figure 13.

C.2 Case study of RoT with iterative traversals

We present examples of the three reasons for RoT performing iterative row-wise table traversals in Figure 14 and Figure 15.

Question
who was the guest judge after kelly brook?

Table

Series	Premiere	Finale	…	Guest judge(s)
One	9 June 2007	17 June 2007		N/A
Two	12 April 2008	31 May 2008		N/A
Three	11 April 2009	30 May 2009	…	Kelly Brook
Four	17 April 2010	5 June 2010		Louis Walsh
Five	16 April 2011	4 June 2011		Louis Walsh
		…		

Output of RoT
To answer the question, I need to figure out who the guest judge was after Kelly Brook based on the given table. …
Let me go through each row one by one.
1. One: The guest judge(s) column is blank. Hmm, maybe it's not applicable or the data isn't provided. I'll skip this one for now.
2. Two: The guest judge(s) column is blank. Same as the first, I'll skip this.
3. Three: The guest judge(s) is Kelly Brook. That's the one I'm looking for, but I need to find the next one.
4. Four: The guest judge(s) is Louis Walsh. That's the next one after Kelly Brook.
…
So the answer is Louis Walsh.
Answer: Louis Walsh

Output of Long CoT
<think>
Okay, so I need to figure out who the guest judge was after Kelly Brook on the show. … Looking at the "Guest judge(s)" column, I see that for Series One, the guest judge is listed as N/A. Series Two also has N/A. Series Three has Kelly Brook as the guest judge. **Series Four onwards, the guest judges are listed as N/A again.**
…
Wait, so Kelly Brook was the guest judge in Series Three. The question is asking who was the guest judge after Kelly Brook. **So, after Series Three, the next series is Series Four, but the guest judge there is N/A. Then Series Five also has N/A** …
</think>
Answer: There was no guest judge after Kelly Brook in the subsequent series.

Figure 11: RoT outperforms Long CoT due to the hallucination issue of Long CoT. The row-wise table traversal process is highlighted in **green**, and the hallucinated reasoning steps are highlighted in **red**.

Question
what city is at the top of the list?

Table

Team	City	Venue	…	Past Season
Aboomoslem	Mashhad	Samen		4th
Bargh Shiraz	Shiraz	Hafezieh		7th
Damash Gilan	Rasht	Sardar		15th
Est. Ahvaz	Ahvaz	Takhti		8th
		…		
Persepolis	Tehran	Azadi		Champion
		…		

Output of RoT
To answer the question, I need to look at the table and find the city that is at the top of the list. Let me go through each row one by one.
The table is sorted alphabetically by team name. I'll start from the top:
1. Aboomoslem, Mashhad: Mashhad is the city.
2. Bargh Shiraz, Shiraz: Shiraz is the city.
3. Damash Gilan, Rasht: Rasht is the city.
4. Est. Ahvaz, Ahvaz: Ahvaz is the city.
…
The first city in the list is Mashhad, which is the city of the team Aboomoslem.
Answer: Mashhad

Output of Long CoT
<think>
Okay, so I need to figure out which city is at the top of the list based on table …
First, I should understand what the question is asking. …**it's likely referring to the ranking from the past season**, as that's a common way to list teams in a league table.
Looking at the "Past Season" column, each team has a ranking. The rankings go from 1st place as the highest to lower ranks. So, the team with 1st place is at the top.
Now, I need to find which team is in 1st place. **Scanning through the list, I see that Persepolis is ranked Champion, which I assume is 1st place. Their city is Tehran. So, the city at the top of the list would be Tehran.**
…Yes, Persepolis is the only team with Champion status, so they are 1st.
</think>
Answer: Tehran

Figure 12: RoT outperforms Long CoT due to the misunderstanding issue of Long CoT. The row-wise table traversal process is highlighted in **green**, and the misunderstanding reasoning steps are highlighted in **red**.

The demonstration of RoT used on WikiTQ and TableBench

Parish	Locality	Parish Priest	Founded	Closed
:—	:—	:—	:—	:—
St Mary	Bacup	Fr Frank Thorpe	1852	—-
Our Immaculate Mother & St Anselm	Whitworth	Fr Frank Thorpe	1860	—-
St Joseph	Stacksteads	—-	1947	2005
St Joseph & St Peter	Newchurch-In-Rossendale	Fr Philip Boast	1915	—-
The Immaculate Conception	Haslingden	Fr Canon John Mackie	1854	—-
St Veronica (Chapel of Ease)	Helmshore	Served from The Immaculate Conception	1959	—-
St James the Less	Rawtenstall	Fr David Lupton, Rural Dean	1828	—-
Question:
what's the number of parishes founded in the 1800s?
Solution:
To answer the question, I need to figure out how many parishes were founded in the 1800s based on the given table. Let me look at the table again. The table has several columns: Parish, Locality, Parish Priest, Founded, and Closed. My task is to count how many parishes were established in the 1800s.

First, I should understand what the "Founded" column represents. It likely indicates the year the parish was established. So, I need to look at each row and check the year under the "Founded" column. If the year is in the 1800s, I'll count that parish.

Let me go through each row one by one.

1. St Mary, Bacup: Founded in 1852. That's in the 1800s, so that's one.
2. Our Immaculate Mother & St Anselm, Whitworth: Founded in 1860. Also in the 1800s, so that's two.
3. St Joseph, Stacksteads: The "Founded" column is blank. Hmm, maybe it's not applicable or the data isn't provided. I'll skip this one for now.
4. St Joseph & St Peter, Newchurch-In-Rossendale: Founded in 1915. That's in the early 1900s, so not in the 1800s.
5. The Immaculate Conception, Haslingden: Founded in 1854. That's another one, so three.
6. St Veronica (Chapel of Ease), Helmshore: Founded in 1959. That's in the 1900s, so not relevant.
7. St James the Less, Rawtenstall: Founded in 1828. That's in the 1800s, so four.

So the answer is four.

Answer: 4

The demonstration of RoT used on HiTab

admissions to adult custody, by sex and by indigenous identity and jurisdiction, 2018/2019
jurisdiction	male	male	male	male	male	male	female	female	female	female	female	female
:—	:—	:—	:—	:—	:—	:—	:—	:—	:—	:—	:—	:—
jurisdiction	indigenous	indigenous	non-indigenous	non-indigenous	unknown indigenous identity	unknown indigenous identity	indigenous	indigenous	non-indigenous	non-indigenous	unknown indigenous identity	unknown indigenous identity
jurisdiction	number	percent change in number from 2017/2018	number	percent change in number from 2017/2018	number	percent change in number from 2017/2018	number	percent change in number from 2017/2018	number	percent change in number from 2017/2018	number	percent change in number from 2017/2018
newfoundland and labrador	333	-7	1,214	-9	0	..	39	-11	187	-4	0	..
prince edward island	25	9	606	11	4	-33	4	-56	129	39	1	0
nova scotia	337	6	3,676	3	18	200	108	21	512	-8	8	14
new brunswick	362	-10	4,319	4	3	0	148	-3	916	17	0	-100
...
Question:
what is the percentage of admissions of non-indigenous males decreased in the provinces and territories compared with the previous year?
Solution:
Okay, so I need to figure out the percentage decrease in admissions of non-indigenous males across the provinces and territories compared to the previous year. Let me look at the table provided.
I need to find the percentage change for non-indigenous males in each province and territory. Let me go through each row:
1. Newfoundland and Labrador: Non-indigenous males have a percent change of -9%.
2. Prince Edward Island: Non-indigenous males have a percent change of 11%.
3. Nova Scotia: Non-indigenous males have a percent change of 3%...
It shows a -7% change for non-indigenous males. I think that's the answer because the total row aggregates all the data, so it's the most accurate representation of the overall percentage change.

Answer: 7

Table 4: The demonstrations of RoT used on three datasets.

The prompt of RoT without Iteration

Your task is to think step by step by traversing the given table to solve the question.
Note that:
1. You must traverse the table row by row once and only once.
2. Represent your answer with: Answer: <Your Answer>.
Here is an example:
...
Based on the above example, you need to traverse the table below and answer the question.

{Table}
Question:
{Question}
Solution:

The prompt of RoT without Traversal

Your task is to think step by step to solve the question.
Note that:
1. You must reflect your reasoning iteratively until you get a correct answer.
2. Represent your answer with: Answer: <Your Answer>.
Here is an example:
...
Based on the above example, you need to answer the question below.

{Table}
Question:
{Question}
Solution:

Table 5: The prompts used in the ablation experiments.

Question
which episode came next after "candy sale"?

Table

Series	Season	Title	...	Original air date
1	1	The Charity		October 15, 1994
2	1	The Practical Joke War		-
		...		
12	1	Candy Sale		January 26, 1995
13	1	The Big Bully		February 2, 1995

Output of RoT
To answer the question, I need to figure out which episode came next after "Candy Sale" based on the given table…
Let me go through each row one by one.
1. Series 1, Season 1, Title: "The Charity", Original air date: October 15, 1994. This is not "Candy Sale".
...
12. Series 1, Season 1, Title: "Candy Sale", Notes: ..., Original air date: January 26, 1995. This is the episode I'm looking for.
13. Series # 1, Season # 1, Title: "The Big Bully", Notes: ..., Original air date: February 2, 1995. This is the next episode after "Candy Sale".
So the answer is "The Big Bully".
Answer: The Big Bully

Output of Long CoT
<think>
Okay, so I need to figure out which episode comes after "Candy Sale" in the show. Let me look at the list provided. The episodes are numbered from 1 to 13, and I need to find the one that comes right after episode 12, which is "Candy Sale."

Looking at the list, episode 12 is titled "Candy Sale" and it's from January 26, 1995. **The next episode after that would be episode 13. So, the answer should be episode 13.**
</think>

Answer: 13

Figure 13: RoT outperforms Long CoT due to the locating error of Long CoT. The row-wise table traversal process is highlighted in **green**, and the incorrect linking reasoning steps are highlighted in **red**.

Question
In the context of the NFL draft, the "pick efficiency" of a college is calculated as the total number of players drafted from that college divided by the total number of players drafted in the round. Based on this definition, which college has the highest pick efficiency in the first 10 rounds of the draft?

Table

round	pick	overall	name	position	college
1	10	10	forest evashevski	rb	michigan
3	10	25	fred davis	ot	alabama
5	10	40	jim stuart	ot	oregon
6	10	50	ed cifers	e	tennessee
7	10	60	al krueger	e	southern california
8	10	70	henry wilder	rb	iowa state
		…			

Output
To determine the college with the highest pick efficiency in the first 10 rounds of the draft, we need to calculate the pick efficiency for each college. …
First, let's identify the colleges with players drafted in the first 10 rounds and the number of players drafted from each:
1. Michigan - 1 player (round 1)
2. Alabama - 2 players (rounds 3, 10)
3. Oregon - 1 player (round 5)
4. Tennessee - 1 player (round 6)
…
Now, let's calculate the pick efficiency for each college in the first 10 rounds:
1. Michigan - 1 player / 8 players = 1/8
2. Alabama - 2 players / 8 players (rounds 3 and 10) = 2/8 = 1/4
3. Oregon - 1 player / 8 players = 1/8
4. Tennessee - 1 player / 8 players = 1/8
…
Alabama's two players in the first 10 rounds give it the highest pick efficiency.
Answer: alabama

Figure 14: RoT performs iterative row-wise table traversals due to Multi-Hop Reasoning. The row-wise table traversal process is highlighted in **green**.

Question
how many episodes in series 10 did fran carroll write?

Table

#	Directed By	Written By	Original Air Date
1	Frank W. Smith	Simon J. Sharkey	September 14, 1997
2	Frank W. Smith	Simon J. Sharkey	September 28, 1997
3	Gerry Poulson	David Humphries	October 5, 1997
4	Gerry Poulson	David Humphries	October 12, 1997
5	John Reardon	Neil McKay	October 26, 1997
6	John Reardon	Neil McKay	November 2, 1997
		…	

Output
To answer the question, I need to find out how many episodes in series 10 …
Let me go through each row one by one.
1. #1: Aired on September 14, 1997. Not in series 10.
2. #2: Aired on September 28, 1997. Not in series 10.
…
18. #18: Aired on February 22, 1998. Not in series 10.
Wait, I made a mistake! I need to look for the episodes written by Fran Carroll, not the episodes in series 10. …
Let me go through each row one by one.
1. #1: Written By Simon J. Sharkey. Not Fran Carroll.
2. #2: Written By Simon J. Sharkey. Not Fran Carroll.
…
18. #18: Written By Simon J. Sharkey. Not Fran Carroll.
So, Fran Carroll wrote only one episode.
Answer: 1

Figure 15: RoT performs iterative row-wise table traversals due to Reflection. The row-wise table traversal process is highlighted in **green**.

Dataset	Method	Perfromance
WikiTQ	Long CoT (zero-shot)	**62.7**
	Long CoT (one-shot)	45.1
HiTab	Long CoT (zero-shot)	**49.7**
	Long CoT (one-shot)	35.4
TableBench	Long CoT (zero-shot)	**34.9**
	Long CoT (one-shot)	25.3

Table 6: Performance of Long CoT using R1-Llama-8B with zero-shot and few-shot.

Method	Accuracy
Llama3-70B	
FlexTaF (Zhang et al., 2024b)	69.9
Llama3.1-70B	
PoTable (Mao et al., 2025)	65.6
SS-CoT (Jiang et al., 2024b)	76.8
TableMaster (Cao, 2025)	78.0
Qwen2-72B	
MACT (Zhou et al., 2025)	72.6
Llama3.3-70B	
Binder (Cheng et al., 2023)	52.2
Dater (Ye et al., 2023)	59.5
Chain-of-Table (Wang et al., 2024)	62.1
Table-Critic (Yu et al., 2025)	70.1
RoT	**78.7**

Table 7: Performance comparison between RoT and previous methods with comparable scale models on WikiTQ.

Method	Accuracy
GPT-3.5	
Zhao et al. (2023)	50.0
code-davinci-002	
Cao et al. (2023)	69.3
Qwen2-72B	
GraphOTTER (Li et al., 2025a)	72.7
Llama3.1-70B	
SS-CoT (Jiang et al., 2024b)	**79.1**
Qwen2.5-32B	
RoT	76.6

Table 8: Performance comparison between RoT and previous methods with comparable scale models on HiTab.

Method	Accuracy
Llama3.1-70B	
Wu et al. (2024)	43.9
Llama3.3-70B	
RoT	**44.8**

Table 9: Performance comparison between RoT and previous methods with comparable scale models on TableBench.

Method/Model	WikiTQ	HiTab	TableBench
RoT	63.6	56.6	35.7
TableLLM-7B (Zhang et al., 2025c)	58.8*	-	-
TableLlama-7B (Zhang et al., 2024a)	35.0	64.7*	-
TableBench LLM-8B (Wu et al., 2024)	-	-	35.3*

Table 10: Comparison of performance between RoT and table-specific LLMs. An asterisk (*) indicates performance on the training set of the corresponding dataset.

Method	Average Traversal Count	Performance
Original prompt	1.10382	**63.6**
Prompt variant 1	1.11096	63.1
Prompt variant 2	1.11786	62.8
Prompt variant 3	1.11098	63.2

Table 11: The average number of traversals for RoT using the original prompt and three paraphrased prompts.

Method	Model	WikiTQ	HiTab	TableBench
RoT	Llama3.1-8B	**628**	**292**	**756**
RoT	R1-Llama-8B	649	471	867
Long CoT	R1-Llama-8B	677	602	1225

Table 12: Comparison of RoT with fixing the traversal count on WikiTQ using LLaMA3.1-8B.